

Groundwater, Soils & Sediments

FE Environmental — Day 6

Study set • 2026-06-22 • 20 questions • every numeric answer recomputed in Python

Handbook fidelity. Per the NCEES FE Reference Handbook v10.6 (Civil Eng. → Groundwater, p. 297–299 and Environmental Eng. → Remediation, p. 330): Darcy’s Law $Q = -KA dh/dx$, the specific discharge $q = -K dh/dx$, seepage velocity $v = q/n$, the constant-head test $K = Q/(iAt_e)$, the falling-head test $K = 2.303(aL)/(At_e) \log_{10}(h_1/h_2)$, transmissivity $T = Kb$, Dupuit’s Formula (unconfined well), and the Thiem Equation (confined well) are all printed in v10.6. The Retardation Factor $R = 1 + (\rho_b/n_e)K_d$, $K_d = K_{oc}f_{oc}$, and Vadose Zone Penetration $D = R_v V/A$ are in the Environmental Engineering section of v10.6. Soil index properties (void ratio, porosity, etc.) appear in the Civil Engineering — Geotechnical section. The contaminant velocity relation $v_c = v/R$ is a direct consequence of the Retardation Factor definition and is flagged as fundamental.

Q1. [D6-Q01 • MCQ • Porosity — definition and calculation]

A soil sample has a total volume of 100 cm^3 and a void volume of 35 cm^3 . The porosity n of the soil is:

Relevant: $n = \frac{V_v}{V_T}$

- (A) 0.25
- (B) **0.35** ✓
- (C) 0.46
- (D) 0.54

Answer: (B)

$$n = \frac{V_v}{V_T} = \frac{35}{100} = 0.35.$$

Handbook: Geotechnical Engineering — Soil Properties • Ctrl-F: Porosity

Q2. [D6-Q02 • MCQ • Void ratio to porosity]

A sandy soil has a void ratio $e = 0.65$. The porosity n is most nearly:

Relevant: $n = \frac{e}{1 + e}$

- (A) **0.39** ✓
- (B) 0.54
- (C) 0.65
- (D) 0.71

Answer: (A)

$$n = \frac{e}{1 + e} = \frac{0.65}{1.65} = 0.394 \approx 0.39.$$

Handbook: Geotechnical Engineering — Soil Properties • Ctrl-F: Void Ratio; Porosity

Q3. [D6-Q03 • MCQ • Confined vs. unconfined aquifer]

An aquifer that is bounded above by an impermeable confining layer (aquitard) so that the groundwater is under pressure greater than atmospheric is called a:

- (A) perched aquifer
- (B) unconfined (water-table) aquifer
- (C) **confined (artesian) aquifer** ✓
- (D) vadose zone

Answer: (C)

A confined (artesian) aquifer is overlain by an impermeable confining layer; water pressure exceeds atmospheric, so water in a tightly cased well rises above the top of the aquifer. An unconfined (water-table) aquifer has a free surface exposed to atmospheric pressure.

Handbook: Groundwater — Well Drawdown • Ctrl-F: Confined aquifer; Artesian

Q4. [D6-Q04 • MCQ • Darcy's Law — discharge rate (SI)]

A saturated sandy formation has a hydraulic conductivity $K = 4.0 \times 10^{-4}$ m/s. The hydraulic gradient across a flow cross-section of $A = 2.0 \text{ m}^2$ is $i = 0.015$. The Darcy discharge rate Q is most nearly:

Relevant: $Q = -KA \frac{dh}{dx} = KiA$ (magnitude)

- (A) $6.0 \times 10^{-3} \text{ m}^3/\text{s}$
- (B) **$1.2 \times 10^{-5} \text{ m}^3/\text{s}$** ✓
- (C) $2.4 \times 10^{-3} \text{ m}^3/\text{s}$
- (D) $3.6 \times 10^{-3} \text{ m}^3/\text{s}$

Answer: (B)

$$Q = KiA = (4.0 \times 10^{-4})(0.015)(2.0) = 1.2 \times 10^{-5} \text{ m}^3/\text{s}.$$

Handbook: Groundwater — Darcy's Law • Ctrl-F: Darcy's Law

Q5. [D6-Q05 • MCQ • Hydraulic gradient]

Two monitoring wells are 120 m apart along the direction of groundwater flow. Water-level elevations are 23.5 m and 21.8 m above the same datum (flow is from the high well to the low well). The hydraulic gradient i is most nearly:

Relevant: $i = \frac{\Delta h}{\Delta x}$

- (A) 0.0085
- (B) 0.0100
- (C) **0.0142 ✓**
- (D) 0.0200

Answer: (C)

$$i = \frac{\Delta h}{\Delta x} = \frac{23.5 - 21.8}{120} = \frac{1.7}{120} = 0.0142.$$

Handbook: Groundwater — Darcy's Law • Ctrl-F: Hydraulic Gradient

Q6. [D6-Q06 • MCQ • Seepage velocity vs. Darcy velocity]

An aquifer has hydraulic conductivity $K = 3.0 \times 10^{-5}$ m/s, hydraulic gradient $i = 0.010$, and effective porosity $n = 0.25$. The average seepage (pore-water) velocity v is most nearly:

Relevant: $q = Ki$ (specific discharge / Darcy velocity) $v = \frac{q}{n} = \frac{Ki}{n}$ (seepage velocity)

- (A) 3.0×10^{-5} m/s
- (B) 7.5×10^{-5} m/s
- (C) **1.2×10^{-6} m/s ✓**
- (D) 1.5×10^{-6} m/s

Answer: (C)

$q = Ki = (3.0 \times 10^{-5})(0.010) = 3.0 \times 10^{-7}$ m/s. $v = q/n = 3.0 \times 10^{-7}/0.25 = 1.2 \times 10^{-6}$ m/s. The seepage velocity is larger than the Darcy velocity because water flows only through the pore space (fraction n of the total area).

Handbook: Groundwater — Darcy's Law • Ctrl-F: Seepage Velocity; Darcy Velocity

Q7. [D6-Q07 • MCQ • Constant-head permeability test]

A constant-head permeability test on a sand sample yields: total discharge $Q = 1.5 \times 10 \text{ m}^3$, elapsed time $t = 60 \text{ s}$, hydraulic gradient $i = 0.80$, specimen cross-sectional area $A = 5 \times 10^3 \text{ m}^2$. The hydraulic conductivity K is most nearly:

Relevant: $K = \frac{Q}{i A t_e}$

- (A) $3.1 \times 10 \text{ m/s}$
- (B) **$6.3 \times 10 \text{ m/s}$** ✓
- (C) $1.3 \times 10 \text{ m/s}$
- (D) $2.5 \times 10 \text{ m/s}$

Answer: (B)

$$K = \frac{Q}{i A t_e} = \frac{1.5 \times 10^{-5}}{(0.80)(5 \times 10^{-3})(60)} = \frac{1.5 \times 10^{-5}}{0.240} = 6.25 \times 10^{-5} \text{ m/s} \approx 6.3 \times 10^{-5} \text{ m/s}.$$

Handbook: Groundwater — Hydraulic Conductivity/Permeability • Ctrl-F: Constant Head; Permeability Test

Q8. [D6-Q08 • MCQ • Transmissivity]

A confined aquifer has a hydraulic conductivity $K = 1.2 \times 10 \text{ m/s}$ and a saturated thickness $b = 25 \text{ m}$. The transmissivity T is most nearly:

Relevant: $T = Kb$

- (A) $1.0 \times 10^3 \text{ m}^2/\text{s}$
- (B) **$3.0 \times 10^3 \text{ m}^2/\text{s}$** ✓
- (C) $4.8 \times 10^3 \text{ m}^2/\text{s}$
- (D) $1.2 \times 10 \text{ m}^2/\text{s}$

Answer: (B)

$$T = Kb = (1.2 \times 10^{-4})(25) = 3.0 \times 10^{-3} \text{ m}^2/\text{s}.$$

Handbook: Groundwater — Transmissivity • Ctrl-F: Transmissivity

Q9. [D6-Q09 • Numeric • Darcy discharge — USCS]

A confined aquifer has a hydraulic conductivity $K = 5 \times 10 \text{ ft/s}$. Two monitoring wells spaced 600 ft apart show a head drop of 4.8 ft . The flow cross-sectional area is $1,200 \text{ ft}^2$. Find the Darcy discharge Q (in ft^3/s).

Relevant: $i = \Delta h / \Delta x$ $Q = KiA$

Numeric entry.

Answer: $\approx 4.8 \times 10 \text{ ft}^3/\text{s}$

$$i = 4.8/600 = 0.0080. \quad Q = KiA = (5 \times 10^{-5})(0.0080)(1200) = 4.8 \times 10^{-4} \text{ ft}^3/\text{s}.$$

Handbook: Groundwater — Darcy's Law • Ctrl-F: Darcy's Law

Q10. [D6-Q10 • MCQ • Dupuit's Formula — unconfined well (USCS)]

An unconfined aquifer well has reached steady-state pumping. Given $K = 1.5 \times 10^{-4}$ ft/s, $h = 22$ ft (saturated height at well, $r = 2$ ft), $h = 32$ ft (at $r = 400$ ft), the steady pumping rate Q is most nearly:

Relevant: $Q = \frac{\pi K(h_2^2 - h_1^2)}{\ln(r_2/r_1)}$

- (A) 0.024 cfs
- (B) 0.035 cfs
- (C) **0.048 cfs ✓**
- (D) 0.072 cfs

Answer: (C)

$$h_2^2 - h_1^2 = 32^2 - 22^2 = 1024 - 484 = 540 \text{ ft}^2. \ln(400/2) = \ln(200) = 5.298. Q = \frac{\pi(1.5 \times 10^{-4})(540)}{5.298} = \frac{0.2545}{5.298} = 0.0480 \text{ cfs.}$$

Handbook: Groundwater — Dupuit's Formula • Ctrl-F: Dupuit; Unconfined Well

Q11. [D6-Q11 • MCQ • Thiem Equation — confined well (SI)]

A confined aquifer ($b = 30$ m, $K = 8 \times 10^{-5}$ m/s) is pumped to steady state. Piezometric heads: $h = 18$ m at $r = 0.2$ m and $h = 25$ m at $r = 200$ m. The pumping rate Q is most nearly:

Relevant: $T = Kb \quad Q = \frac{2\pi T(h_2 - h_1)}{\ln(r_2/r_1)}$

- (A) $7.5 \times 10^3 \text{ m}^3/\text{s}$
- (B) **$1.5 \times 10^2 \text{ m}^3/\text{s} \checkmark$**
- (C) $2.3 \times 10^2 \text{ m}^3/\text{s}$
- (D) $3.1 \times 10^2 \text{ m}^3/\text{s}$

Answer: (B)

$$T = Kb = (8 \times 10^{-5})(30) = 2.4 \times 10^{-3} \text{ m}^2/\text{s}. \ln(200/0.2) = \ln(1000) = 6.908. Q = \frac{2\pi(2.4 \times 10^{-3})(25 - 18)}{6.908} = \frac{2\pi(0.0168)}{6.908} = \frac{0.1056}{6.908} = 0.01528 \text{ m}^3/\text{s} \approx 1.5 \times 10^{-2} \text{ m}^3/\text{s}.$$

Handbook: Groundwater — Thiem Equation • Ctrl-F: Thiem; Confined Well

Q12. [D6-Q12 • MCQ • Specific capacity of a well]

A municipal well is pumped at a rate $Q = 0.050$ cfs and the resulting drawdown at the well is $D_w = 8.0$ ft. The specific capacity of the well is:

Relevant: $SC = \frac{Q}{D_w}$

- (A) 0.0031 cfs/ft
- (B) **0.0063 cfs/ft** ✓
- (C) 0.0125 cfs/ft
- (D) 0.40 cfs/ft

Answer: (B)

$$SC = Q/D_w = 0.050/8.0 = 0.00625 \text{ cfs/ft} \approx 0.0063 \text{ cfs/ft.}$$

Handbook: Groundwater — Dupuit's Formula (specific capacity) • Ctrl-F: Specific Capacity; Drawdown

Q13. [D6-Q13 • MCQ • Falling-head permeability test]

A falling-head permeability test on a clay sample gives: standpipe area $a = 1 \text{ cm}^2$, specimen cross-sectional area $A = 50 \text{ cm}^2$, specimen length $L = 15 \text{ cm}$, elapsed time $t = 480 \text{ s}$, initial head $h = 80 \text{ cm}$, final head $h = 40 \text{ cm}$. The hydraulic conductivity K is most nearly:

Relevant: $K = 2.303 \frac{aL}{A t_e} \log_{10} \left(\frac{h_1}{h_2} \right)$

- (A) $2.2 \times 10 \text{ m/s}$
- (B) **$4.3 \times 10 \text{ m/s}$** ✓
- (C) $8.7 \times 10 \text{ m/s}$
- (D) $1.4 \times 10 \text{ m/s}$

Answer: (B)

$$K = 2.303 \frac{(1)(15)}{(50)(480)} \log_{10}(80/40) = 2.303 \times 6.25 \times 10^{-4} \times 0.3010 = 4.33 \times 10^{-4} \text{ cm/s} = 4.3 \times 10^{-6} \text{ m/s.}$$

Handbook: Groundwater — Hydraulic Conductivity/Permeability • Ctrl-F: Falling Head; Permeability Test

Q14. [D6-Q14 • Numeric • Dupuit's Formula — pumping rate in gpm (USCS)]

An unconfined aquifer well reaches steady state with the following data: $K = 2.0 \times 10 \text{ ft/s}$, well radius $r = 1.5 \text{ ft}$ with saturated height $h = 30 \text{ ft}$, observation well at $r = 600 \text{ ft}$ with $h = 45 \text{ ft}$. Find the steady-state pumping rate Q in gallons per minute ($1 \text{ ft}^3/\text{s} = 448.8 \text{ gpm}$).

$$\text{Relevant: } Q = \frac{\pi K(h_2^2 - h_1^2)}{\ln(r_2/r_1)}$$

Numeric entry.

Answer: $\approx 53 \text{ gpm}$

$$h_2^2 - h_1^2 = 45^2 - 30^2 = 2025 - 900 = 1125 \text{ ft}^2. \quad \ln(600/1.5) = \ln(400) = 5.991. \quad Q = \frac{\pi(2.0 \times 10^{-4})(1125)}{5.991} = \frac{0.7069}{5.991} = 0.1180 \text{ ft}^3/\text{s}. \quad 0.1180 \times 448.8 = 52.9 \approx 53 \text{ gpm}.$$

Handbook: Groundwater — Dupuit's Formula • Ctrl-F: Dupuit; Unconfined Well

Q15. [D6-Q15 • Numeric • Thiem Equation — pumping rate (SI)]

A confined aquifer has transmissivity $T = 3.5 \times 10^3 \text{ m}^2/\text{s}$. At steady state, piezometric heads are $h = 12 \text{ m}$ at $r = 0.25 \text{ m}$ (well face) and $h = 18 \text{ m}$ at $r = 250 \text{ m}$. Determine the pumping rate Q (in m^3/s).

$$\text{Relevant: } Q = \frac{2\pi T(h_2 - h_1)}{\ln(r_2/r_1)}$$

Numeric entry.

Answer: $\approx 0.0191 \text{ m}^3/\text{s}$

$$\ln(250/0.25) = \ln(1000) = 6.908. \quad Q = \frac{2\pi(3.5 \times 10^{-3})(18 - 12)}{6.908} = \frac{2\pi(0.021)}{6.908} = \frac{0.13195}{6.908} = 0.01910 \text{ m}^3/\text{s}.$$

Handbook: Groundwater — Thiem Equation • Ctrl-F: Thiem; Confined Well

Q16. [D6-Q16 • MCQ • Retardation factor — contaminant velocity]

A contaminant in an aquifer has a retardation factor $R = 4.0$. If the average seepage velocity of the groundwater is 0.50 m/day , the velocity of the contaminant plume is most nearly:

$$\text{Relevant: } v_c = \frac{v}{R} \text{ (follows directly from } R = 1 + (\rho_b/n_e)K_d)$$

- (A) **0.125 m/day** ✓
- (B) 0.250 m/day
- (C) 0.500 m/day
- (D) 2.00 m/day

Answer: (A)

$v_c = v/R = 0.50/4.0 = 0.125 \text{ m/day}$. The retardation factor R expresses how much slower a sorbing contaminant travels relative to the average pore-water velocity. $R = 1$ means no retardation (conservative tracer).

Handbook: Remediation — Retardation Factor • Ctrl-F: Retardation Factor

Q17. [D6-Q17 • MCQ • Distribution coefficient K_d from K_{oc} and f_{oc}]

A soil has a fraction of organic carbon $f_{oc} = 0.02$ and an organic carbon partition coefficient $K_{oc} = 150$ mL/g. The distribution coefficient K_d is most nearly:

Relevant: $K_d = K_{oc} f_{oc}$

- (A) 0.013 mL/g
- (B) 0.30 mL/g
- (C) **3.0 mL/g** ✓
- (D) 30 mL/g

Answer: (C)

$$K_d = K_{oc} f_{oc} = 150 \times 0.02 = 3.0 \text{ mL/g.}$$

Handbook: Remediation — Soil-Water Partition Coefficient / Retardation Factor • Ctrl-F: K_{oc} ; Partition Coefficient

Q18. [D6-Q18 • MCQ • Vadose zone penetration depth]

A 0.5 m³ gasoline spill occurs over a surface area of 4 m² at a site with medium to fine sand ($R_v = 80$). Using the Handbook vadose zone penetration formula, the maximum depth of penetration D is most nearly:

Relevant: $D = \frac{R_v V}{A}$

- (A) 5 m
- (B) **10 m** ✓
- (C) 20 m
- (D) 40 m

Answer: (B)

$$D = \frac{R_v V}{A} = \frac{80 \times 0.5}{4} = \frac{40}{4} = 10 \text{ m.}$$

Handbook: Remediation — Vadose Zone Penetration • Ctrl-F: Vadose Zone; Penetration

Q19. [D6-Q19 • Numeric • Retardation factor — full calculation]

An aquifer has a bulk density $\rho_b = 1.8$ g/cm³, effective porosity $n_e = 0.30$, organic carbon partition coefficient $K_{oc} = 200$ cm³/g, and soil organic carbon fraction $f_{oc} = 0.01$. Compute the retardation factor R .

Relevant: $K_d = K_{oc} f_{oc}$ $R = 1 + \frac{\rho_b}{n_e} K_d$

Numeric entry.

Answer: $R = 13$

$K_d = K_{oc} f_{oc} = 200 \times 0.01 = 2.0$ cm³/g. $R = 1 + \frac{\rho_b}{n_e} K_d = 1 + \frac{1.8}{0.30} \times 2.0 = 1 + 6.0 \times 2.0 = 1 + 12 = 13$. This means the contaminant travels at 1/13 the groundwater seepage velocity.

Handbook: Remediation — Retardation Factor • Ctrl-F: Retardation Factor; K_{oc}

Q20. [D6-Q20 • MCQ • Effective porosity — physical meaning]

In the groundwater seepage velocity equation $v = q/n$, the value n is the *effective* porosity rather than total porosity because it:

Relevant: $v = \frac{q}{n} = \frac{Ki}{n}$

- (A) includes all pore space, even that occupied by bound or trapped water
- (B) **excludes pore space occupied by immobile (bound/trapped) water that does not transmit flow ✓**
- (C) is always larger than total porosity in fine-grained soils
- (D) equals the void ratio e for all soil types

Answer: (B)

Effective porosity is the fraction of pore space through which water actually flows — it excludes isolated pores, dead-end pores, and water held by surface tension or adsorption. Because effective porosity $\downarrow$ total porosity, the seepage velocity $v = q/n$ is *higher* than the Darcy velocity q ; it better represents the true speed at which dissolved contaminants (non-retarded) travel.

Handbook: Groundwater — Darcy's Law • Ctrl-F: Effective Porosity; Seepage Velocity